

[ARTICLE TITLE]

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Abstract—The Internet of Things (IoT) spans diverse sectors but faces challenges due to technological heterogeneity and costly deployments. This work presents SimulateIoT, a Model-Driven Development (MDD) methodology and toolset for designing, generating, and executing IoT simulations using high-level domain concepts and distributed microservice deployments. The SimulateIoT metamodel formalizes key IoT elements, including Edge, Fog, and Cloud nodes, sensors, actuators, publish-subscribe topics, synthetic and file-based data sources, Complex Event Processing (CEP) engines, NoSQL storage, notifications, and real-time rules. Models created with an EMF/GMF graphical editor and validated by OCL constraints are transformed via Acceleo into Java Thorntail microservices, Mosquitto MQTT brokers, MongoDB databases, CEP engines (Esper, WSO2), and REST APIs. Simulations are orchestrated through Docker Swarm for distributed execution. Two case studies—a smart building and an agricultural environment—demonstrate the platform’s ability to simulate sensor-actuator interactions, CEP rules, data persistence, and system orchestration. The system offers detailed monitoring of CPU, memory, and network resources. SimulateIoT enables scalable, technology-agnostic IoT modeling that streamlines system design, validation, and deployment compared to ad-hoc approaches. Limitations include partial node mobility support, a fixed domain-specific language, simplified hardware abstractions, and static configuration models; planned enhancements aim to introduce dynamic adaptive behaviors. This framework provides a robust foundation for both professional and educational IoT simulation needs.

Index Terms—[KEYWORD 1], [KEYWORD 2], [KEYWORD 3]

I. INTRODUCTION

The Internet of Things (IoT) is increasingly applied across various domains such as smart cities, smart homes, agriculture, and industrial environments. However, developing, deploying, and testing IoT projects require significant investments in hardware components like devices, fog nodes, cloud nodes, and analytics infrastructure, as well as substantial software development efforts. To mitigate these costs, simulation environments for IoT systems are essential as they facilitate system modeling, reasoning, and optimization prior to actual deployment.

Existing IoT simulation environments predominantly focus on low-level aspects such as network protocols, sensor hardware, and motes, demanding advanced technical knowledge and programming skills from users. These tools often lack high-level abstractions for IoT concepts, such as Edge, Fog, and Cloud layers, and do not provide intuitive modeling or deployment mechanisms. Therefore, there is a pressing need for approaches that enable the design and execution of complex IoT simulations using domain-specific abstractions—including sensors, actuators, heterogeneous nodes, publish-subscribe communication, microservices, and containerization

technologies—all within a user-friendly model-driven development framework.

Several existing approaches have addressed aspects of IoT simulation but exhibit notable limitations that motivate further research. Alwasel et al. (2020) [VERIFICAR] simulate Big Data and software-defined networks in cloud environments but lack graphical interfaces and validation mechanisms for pre-configuration of IoT simulation environments. Levis et al. (2003) [1] with Tossim focus on low-level simulation of TinyOS nodes without defining IoT communication patterns or modeling high-level processes, limiting the scope of IoT system representation. Mehdi et al. (2014) [2] propose Cup-Carbon, which allows graphical simulation of IoT contexts but does not support modeling of storage or complex protocols such as publish/subscribe. Patel and Cassou (2015) offer IoTSuite to facilitate IoT application development, yet it does not encompass simulation capabilities nor modeling of storage and complex processing. Finally, Zeng et al. (2017) [3] present IOTSim, a cloud IoT application simulator focused on big data processing, but without tools for high-level graphical modeling of entire IoT architectures.

In summary, these works generally lack a comprehensive high-level domain-specific language with validation support, comprehensive modeling across the entire IoT architecture layers (Edge, Fog, Cloud), integrated support for publish-subscribe communication protocols, and deployment mechanisms leveraging containerized microservices. These gaps highlight the need for solutions that unify high-level modeling, validation, flexible communication abstractions, and realistic deployment strategies in IoT simulation.

The central proposal of this work is a model-driven development approach to define, generate code for, and deploy IoT simulations. This approach is based on a domain-specific metamodel named SimulateIoT, which provides high-level abstractions representing sensors, actuators, and heterogeneous nodes across Edge, Fog, and Cloud layers. A graphical concrete syntax facilitates intuitive visual modeling, while Object Constraint Language (OCL) rules ensure model validation. From these validated models, model-to-text transformations automatically generate all necessary components, including microservices, MQTT brokers, NoSQL databases, complex event processing engines, and Docker containers for deployment.

This comprehensive framework explicitly addresses the gaps found in prior work by unifying the full IoT architectural stack within a single, validated modeling language that supports publish-subscribe communication patterns and leverages containerized microservice deployment. By doing so, it enables users to efficiently design complex IoT simulation environments and execute them realistically without manual coding, thus overcoming limitations related to low-level focus, lack

of high-level modeling support, and inadequate deployment automation.

- Development of a comprehensive metamodel for IoT simulation that encompasses devices, nodes, communication protocols, storage, and processing capabilities (Sections III and IV).
- Creation of a graphical concrete syntax that enables the design of simulation models conforming to the metamodel and supports OCL-based validation to ensure model correctness (Section IV-B).
- Implementation of a model-to-text transformation workflow that automates the generation of all required code and deployment artefacts for the simulation environment (Section IV-C).
- Integration with industrial technologies, including Docker containers, MQTT communication brokers, and complex event processing engines such as Esper and WSO2, facilitating realistic simulation and seamless deployment (Section IV-D).
- Presentation of two detailed case studies—one in a smart building environment and another in an agricultural setting—that demonstrate the expressiveness and applicability of the proposed approach (Section V).
- Development and deployment of a fully operational simulation platform featuring monitoring and real-time data analysis capabilities, allowing users to observe and analyze system behavior during execution (Sections V and VI).

The validation of this work is conducted through two comprehensive case studies situated in distinct IoT domains: a smart building environment and an agricultural setting. These case studies demonstrate the modeling, code generation, deployment, and execution phases using Docker technology, showcasing the realistic simulation of complex IoT systems composed of various sensors, actuators, and heterogeneous nodes spanning Edge, Fog, and Cloud layers. The evaluation highlights the system's capability for real-time data analysis, integrated monitoring, event management, and rule-based actions, thereby substantiating the expressiveness and practical applicability of the proposed model-driven development approach.

The remainder of the article is organized into clearly differentiated main sections that guide the reader from the motivation behind the work through related research, the proposed SimulateIoT methodology, the design and implementation of supporting tools, detailed case studies focused on smart building and agricultural environments, and culminating with discussions on limitations, target users, and technologies, followed by conclusions and future work. This structured organization facilitates a comprehensive understanding of the development and application of the presented IoT simulation framework.

II. [SECTION 2 TITLE]

[Section 2 content.]

III. [SECTION 3 TITLE]

[Section 3 content.]

IV. [SECTION 4 TITLE]

[Section 4 content.]

V. CONCLUSION

[Conclusions.]

APPENDIX A

APPENDIX

[Appendix content.]

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[FIRST AUTHOR] [Brief author biography.]

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